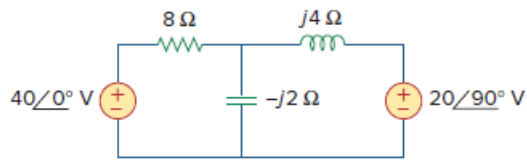


Problem

Calculate the average power absorbed by each of the five elements in the circuit of Fig. 11.6.

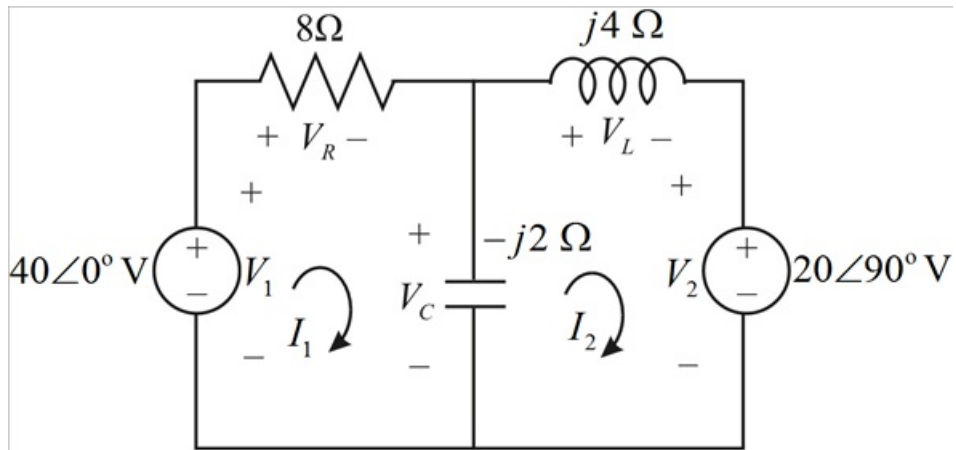
Figure 11.6:



Step-by-step solution

Step 1 of 11

The given circuit is:



Step 2 of 11

Writing mesh analysis in loop (1), we get

$$-40\angle 0^\circ + 8I_1 + (-j2)(I_1 - I_2) = 0$$

$$(8 - j2)I_1 + j2I_2 = 40\angle 0^\circ$$

$$(4 - j1)I_1 + j1I_2 = 20 \quad \text{..... (1)}$$

Writing mesh analysis in loop (2), we get

$$-(j2)(I_2 - I_1) + j4(I_2) + 20\angle 90^\circ = 0$$

$$j2I_1 + j2I_2 = -20\angle 90^\circ$$

$$j1I_1 + j1I_2 = -j10$$

$$I_1 + I_2 = -10$$

$$I_2 = -10 - I_1 \quad \text{..... (2)}$$

Step 3 of 11

Substituting I_2 from equation (2) into equation (1), we get:

$$(4 - j1)I_1 + j1(-10 - I_1) = 20$$

$$I_1(4 - j1 - j1) = 20 + j10$$

$$I_1 = \frac{20 + j10}{4 - j2}$$

$$I_1 = \frac{10}{2} \times \frac{2 + j1}{2 - j1}$$

Step 4 of 11

Rationalizing the denominator, we get:

$$I_1 = 5 \times \frac{2 + j1}{2 - j1} \times \frac{2 + j1}{2 + j1}$$

$$I_1 = 5 \times \frac{3 + j4}{5}$$

$$I_1 = 3 + j4$$

$$I_1 = 5 \angle 53.130^\circ \text{ A}$$

Step 5 of 11

Substituting I_1 value in equation (2), we get:

$$I_2 = -10 - I_1$$

$$I_2 = -10 - (3 + j4)$$

$$I_2 = -13 - j4$$

$$I_2 = 13.601 \angle -162.897^\circ \text{ A}$$

Step 6 of 11

For the voltage source V_1 , the current flowing from it is $-I_1 = -5 \angle 53.130^\circ \text{ A}$ and the voltage across it is $40 \angle 0^\circ \text{ V}$, so that the average power is

$$P_{V_1} = \frac{1}{2} V_{1m} I_{1m} \cos(\theta_v - \theta_i)$$

$$P_{V_1} = \frac{1}{2} \times 40 \times (-5) \times \cos(0^\circ - 53.130^\circ)$$

$$P_{V_1} = -100 \times \cos(-53.130^\circ)$$

$$P_{V_1} = -100 \times 0.6$$

$$\therefore \boxed{P_{V_1} = -60 \text{ W}}$$

It is negative according to the passive sign convention, meaning that the voltage source is supplying power to the circuit.

Step 7 of 11

For the voltage source V_2 , the current flowing from it is $I_2 = 13.601\angle -162.897^\circ$ A and the voltage across it is $20\angle 90^\circ$ V, so that the average power is

$$P_{V_2} = \frac{1}{2} V_{2m} I_{2m} \cos(\theta_v - \theta_i)$$

$$P_{V_2} = \frac{1}{2} \times 20 \times 13.601 \times \cos(90^\circ - (-162.897^\circ))$$

$$P_{V_2} = 136.01 \times \cos(252.897^\circ)$$

$$P_{V_2} = 136.01 \times (-0.2941)$$

$$\therefore \boxed{P_{V_2} = -40 \text{ W}}$$

It is negative according to the passive sign convention, meaning that the voltage source is supplying power to the circuit.

Step 8 of 11

For the resistor, the current through it is $I_1 = 5\angle 53.130^\circ$ A and the voltage across it is

$$V_R = 8I_1$$

$$V_R = 8 \times 5\angle 53.130^\circ$$

$$V_R = 40\angle 53.130^\circ \text{ V}$$

The average power absorbed by 8Ω resistor is

$$P_R = \frac{1}{2} V_{Rm} I_{1m} \cos(\theta_v - \theta_i)$$

$$P_R = \frac{1}{2} \times 40 \times 5 \times \cos(53.130^\circ - 53.130^\circ)$$

$$P_R = 100 \times \cos(0^\circ)$$

$$P_R = 100 \times 1$$

$$\therefore \boxed{P_R = 100 \text{ W}}$$

Step 9 of 11

For the inductor, the current through it is $I_2 = 13.601\angle -162.897^\circ$ A and the voltage across it is $V_L = j4.I_2$

$$V_L = (4\angle 90^\circ) \times (13.601\angle -162.897^\circ)$$

$$V_L = 54.404\angle -72.897^\circ \text{ V}$$

Hence, the average power absorbed by the inductor is

$$P_L = \frac{1}{2} V_{Lm} I_{2m} \cos(\theta_v - \theta_i)$$

$$P_L = \frac{1}{2} \times 54.404 \times 13.601 \times \cos(-72.897^\circ - (-162.897^\circ))$$

$$P_L = 1186.03 \times \cos(90^\circ)$$

$$P_L = 1186.03 \times 0$$

$$\therefore \boxed{P_L = 0 \text{ W}}$$

Step 10 of 11

For the capacitor, the current through it is $I_C = I_1 - I_2$

$$I_C = (3 + j4) - (-13 - j4)$$

$$I_C = 16 + j8$$

$$I_C = 17.89 \angle 26.565^\circ \text{ A}$$

And the voltage across it is

$$V_C = (-j2)I_C$$

$$V_C = (2 \angle -90^\circ)(17.89 \angle 26.565^\circ)$$

$$V_C = 35.78 \angle -63.435^\circ \text{ V}$$

Step 11 of 11

Thus, the average power absorbed by the capacitor is

$$P_C = \frac{1}{2} V_{Cm} I_{Cm} \cos(\theta_v - \theta_i)$$

$$P_C = \frac{1}{2} \times 35.78 \times 17.89 \times \cos(-63.435^\circ - 26.565^\circ)$$

$$P_C = 320.05 \times \cos(-90^\circ)$$

$$P_C = 320.05 \times 0$$

$$\therefore \boxed{P_C = 0 \text{ W}}$$